1. //WAP to input any integer number and check it either it is prime or not.

```
#include <stdio.h>


int main(){
    int num, factor=0;
    printf("Enter an integer number: ");
    scanf("%d", &num);
    for (int i=1; i<=num;i++){
        if (num %i==0)
            factor++;
    }

    if(num==1)
        printf("Neither Prime nor Composite.");

    else if (factor==2)
        printf("The number is prime.");
    else
        printf("The number is not prime.");

        return 0;
}
```

---

2. // WAP to input an integer number n and print all prime numbers from 1 upto n.

```
#include <stdio.h>

int main(){
    int n, factor;
    printf("Enter an integer number: ");
    scanf("%d", &n);
    printf("Prime numbers from 1 to %d are:\n", n);

    for (int num = 2; num <= n; num++) {
        factor = 0;
        for (int i = 1; i <= num; i++) {
            if (num % i == 0)
                factor++;
        }
        if (factor == 2) {
            printf("%d \t", num);
        }
    }
    printf("\n");
    return 0;
}
```

---

3. /*WAP to input an integer number n and calculate the sum of the following series
- 1! + 2! + 3!+ …………….. +n!
- 1+2^2+3^3+…………..+n^n
- 1/1!+2/2!+3/3!+…………..+n/n!*/

```
#include <stdio.h>
#include <math.h>
int main(){
    int n;
    long long fact, sum1 = 0, sum2 = 0;
    double sum3 = 0.0;

    printf("Enter an integer number: ");
    scanf("%d", &n);

    for (int i = 1; i <= n; i++) {
        // Calculate factorial
        fact = 1;
        for (int j = 1; j <= i; j++) {
            fact *= j;
        }
        sum1 += fact;               // Sum for 1! + 2! + ... + n!
        sum2 += pow(i, i);          // Sum for 1 + 2^2 + ... + n^n
        sum3 += (double)i / fact;   // Sum for 1/1! + 2/2! + ... + n/n!
    }

    for(int i=1; i<=n;i++){
        printf("+%d! ",i);
    }
    printf("= %lld\n", sum1);
    for (int i=1; i<=n;i++){
        printf("(%d^%d) + ",i,i);
    }
    printf(" = %lld\n", sum2);
    for (int i=1; i<=n;i++){
        printf("(%d/%d!) + ",i,i);
    }
    printf("= %.9lf\n", sum3);

    return 0;
}
```

---

4. //WAP to input five integer numbers into an array and print them.

```
#include <stdio.h>

int main(){
    int a[5];
    printf("Enter five integer numbers:\n");
    for (int i=0; i<5; i++){
        printf("%d) ",i+1);
        scanf("%d", &a[i]);
    }
    printf("The numbers are:\n");
    for (int i=0; i<5; i++){
        printf("%d\t", a[i]);
    }
}
```

---

5. /*WAP to input 10 integer numbers into an array and the sum and
average of all array elements.*/

```
#include <stdio.h>

int main(){
    int arr[10], sum = 0;
    printf("Enter 10 integer numbers:\n");
    for (int i = 0; i < 10; i++) {
        printf("%d) ", i + 1);
        scanf("%d", &arr[i]);
        sum += arr[i];
    }
    printf("The sum of all array elements is: %d\n", sum);
    printf("The average of all array elements is: %.2f\n", sum / 10.0);
    return 0;
}
```

6.// WAP to input 10 float numbers into an array and display them in reverse order

```
#include <stdio.h>

int main(){
    float arr[10];
    printf("Enter 10 float numbers:\n");
    for (int i = 0; i < 10; i++) {
        printf("%d) ", i + 1);
        scanf("%f", &arr[i]);
    }
    printf("The numbers in reverse order are:\n");
    for (int i = 9; i >= 0; i--) {
        printf("%.2f\t", arr[i]);
    }
    return 0;
}
```

7.// Write a program to count the number of even and odd elements in an integer array of size 10.

```
#include <stdio.h>

int main(){
    int arr[10], even = 0, odd = 0;
    printf("Enter 10 integer numbers:\n");
    for (int i = 0; i < 10; i++) {
        printf("%d) ", i + 1);
        scanf("%d", &arr[i]);
        if (arr[i] % 2 == 0) {
            even++;
        } else {
            odd++;
        }
    }
    printf("Number of even elements: %d\n", even);
    printf("Number of odd elements: %d\n", odd);
    return 0;
}
```

---

8. //WAP to input 10 float numbers into an array calculate the sum of those numbers that are divisible by either 3 or 5.

```
#include <stdio.h>
int main(){
    float arr[10], sum = 0.0;
    printf("Enter 10 float numbers:\n");
    for (int i = 0; i < 10; i++) {
        printf("%d) ", i + 1);
        scanf("%f", &arr[i]);
        if (((int)arr[i] % 3 == 0) || ((int)arr[i] % 5 == 0)) {
            sum += arr[i];
        }
    }
    for (int i = 0; i < 10; i++) {
        if (((int)arr[i] % 3 == 0) || ((int)arr[i] % 5 == 0)) {
            printf(" +%.2f ",arr[i]);
        }
    }
    printf("= %.2f\n", sum);
    return 0;
}
```

---

9. //WAP to input 5 numbers into an array and find the largest number.

```
#include <stdio.h>

int main(){
    int arr[5], largest;
```

```
    printf("Enter 5 numbers:\n");
    for (int i = 0; i < 5; i++) {
        printf("%d) ", i + 1);
        scanf("%d", &arr[i]);
    }
    largest = arr[0];
    for (int i = 1; i < 5; i++) {
        if (arr[i] > largest) {
            largest = arr[i];
        }
    }
    printf("The largest number is: %d\n", largest);
    return 0;
}
```

---

10. // WAP to input 10 numbers into an array and calculate the sum of odd numbers only.

```
#include <stdio.h>

int main(){
    int arr[10], sum = 0;
    printf("Enter 10 numbers:\n");
    for (int i = 0; i < 10; i++) {
        printf("%d) ", i + 1);
        scanf("%d", &arr[i]);
        if (arr[i] % 2 != 0) {
            sum += arr[i];
        }
    }
    for (int i = 0; i < 10; i++) {
        if (arr[i] % 2 != 0) {
            printf(" +%d ",arr[i]);
        }
    }
    printf("= %d\n", sum);
    return 0;
}
```

---

11. // WAP to input 10 integer numbers into an array and separate positive and negative numbers into two different arrays and finally print them.

```
#include <stdio.h>

int main(){
    int arr[10], pos[10], neg[10];
    int posCount = 0, negCount = 0;

    printf("Enter 10 integer numbers:\n");
    for (int i = 0; i < 10; i++) {
        printf("%d) ", i + 1);
        scanf("%d", &arr[i]);
        if (arr[i] >= 0) {
```

```
            pos[posCount++] = arr[i];
        } else {
            neg[negCount++] = arr[i];
        }
    }

    printf("Positive numbers:\n");
    for (int i = 0; i < posCount; i++) {
        printf("%d ", pos[i]);
    }
    printf("\n");

    printf("Negative numbers:\n");
    for (int i = 0; i < negCount; i++) {
        printf("%d ", neg[i]);
    }
    printf("\n");

    return 0;
}
```

---

12. //WAP to input 10 integer numbers into an array, input another number and search weather that number is in the array or not.

```
#include <stdio.h>

int main(){
    int arr[10], search;
    printf("Enter 10 integer numbers:\n");
    for (int i = 0; i < 10; i++) {
        printf("%d) ", i + 1);
        scanf("%d", &arr[i]);
    }
    printf("Enter a number to search: ");
    scanf("%d", &search);
    int index=0;

    for (int i = 0; i < 10; i++) {
       printf("Checking progress: %d/10\n", i+1);
       if (arr[i] == search) {
           index = i + 1;
    }
}
    if (index != 0) {
        printf("Number %d found in the array at position %d index=arr[%d].\n", search, index, index - 1);
        } else {
        printf("Number %d not found in the array.\n", search);
        }

    return 0;
}
```

There is another version of Q12 at:
"https://github.com/Not-Yug-Sharma/Notes-Obsidian/blob/main/SEM%20I/Notes/CSIT%20115-%20%20Programming%20in%20C/Programs/Lab3/q12v1.c"
*****THIS HAS BEEN DONE WITH SUPPORT FROM GEMINI-AI FOR FUNCTIONS under <windows.h>*****